CRANKSHAFT

DESCRIPTION

The crankshaft is made of nodular cast iron and includes five main bearing journals and four connecting rod journals (Fig. 41). The number three journal is the location for the thrust bearing. The mains and connecting rod journals have undercut fillet radiuses that are rolled for added strength. To optimize bearing loading, eight counterweights are used.

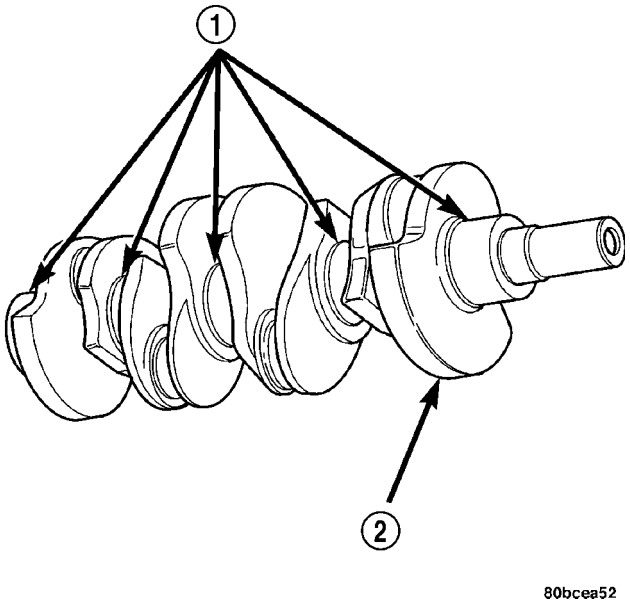

Fig. 41 Crankshaft - Typical

1 - MAIN BEARING JOURNALS
2 - COUNTER BALANCE WEIGHTS

OPERATION

The crankshaft transfers force generated by combustion within the cylinder to the flywheel or flexplate.

STANDARD PROCEDURE - CRANKSHAFT END PLAY

(1) Using Dial Indicator C-3339 and Mounting Post L-4438, attach to front of engine, locating probe perpendicular on nose of crankshaft (Fig. 42).

(2) Move crankshaft all the way to the rear of its travel.

(3) Zero the dial indicator.

(4) Move crankshaft all the way to the front and read the dial indicator. Refer to Engine Specifications.

REMOVAL

NOTE: Crankshaft can not be removed when engine is in vehicle.

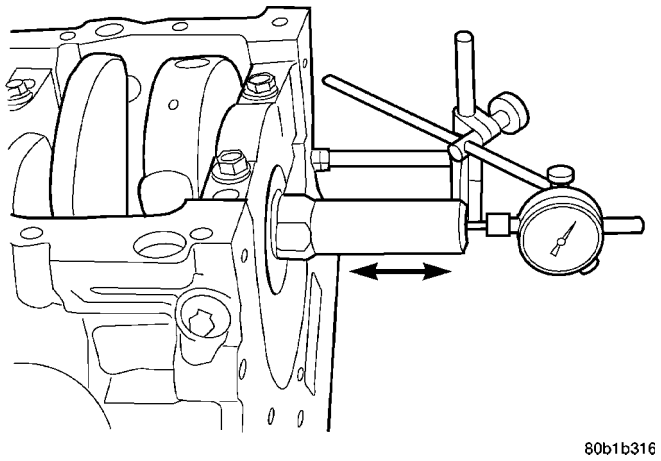

Fig. 42 CHECKING CRANKSHAFT END PLAY

(1) Remove engine assembly from vehicle. (Refer to 9 - ENGINE - REMOVAL)

(2) Remove flex plate and crankshaft rear oil seal.

(3) Mount engine on a repair stand.

(4) Drain engine oil and remove oil filter.

(5) Remove the oil pan. (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - REMOVAL)

(6) Remove the timing belt covers. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - REMOVAL)

(7) Remove the timing belt. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - REMOVAL)

(8) Remove the oil pump. (Refer to 9 - ENGINE/LUBRICATION/OIL PUMP - REMOVAL)

(9) Remove balance shafts and housing assembly. (Refer to 9 - ENGINE/VALVE TIMING/BALANCE SHAFT - REMOVAL)

(10) Using a permanent ink or paint marker, identify cylinder number on each connecting rod cap (Fig. 43).

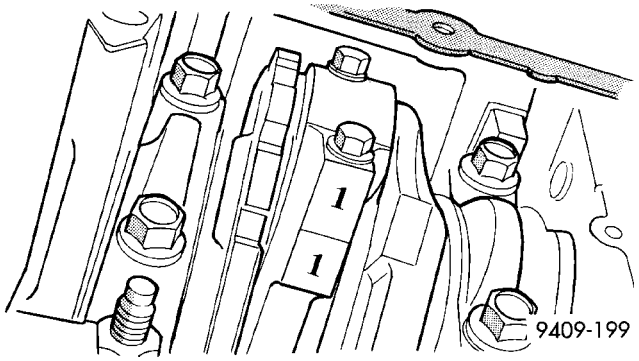

Fig. 43 Identify Connecting Rod to Cylinder-Typical

CAUTION: DO NOT use a number stamp or a punch to mark connecting rods. Damage to connecting rod could occur.